WHAT'S IN A NAME?

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RNA splicing is a necessary step in eukaryotic gene expression. In humans, it's estimated that around 90% of genes are alternatively spliced and a growing body of evidence points to alternative splicing as a major form of cellular control. In recent years, the field of alternative splicing has expanded rapidly due to:

- Better bioinformatics tools to analyse short-read RNA-Seq[1,2]
- The advent of long-read RNA-Seq[3]
- Diseases and cancer associated with dysregulated splicing, and
- Therapeutic drugs that artificially modulate splicing (esp. antisense oligonucleotides)[4,5]

Despite its importance, there is no standardised nomenclature to describe the structure of spliced RNA!

Too confusing!

Literature descriptions of transcripts and splicing events are inconsistent and irreproducible.

Currently the most common ways to describe splicing are:

- Splice modes (cassette, A3SS, A5SS, IR...)
- Ad-hoc numbering for isoforms
- Transcript schematics and sashimi plots
- Ensembl, RefSeq or LRG identifiers, and
- Genomic co-ordinates

Each format has its drawbacks in regards to **readability**, **conciseness**, **reproducibility** and **ease of data mining**.

Better names

To address the need for better names, we combined many existing naming conventions into a unified shorthand system. We call it the **Extended Delta Notation (EDN)**.*

The EDN describes splicing relative to reference transcripts by using modifiers for exons, introns and junctions – a more natural way of defining alternative splicing because exon boundaries only occur at certain points.† Any splice event can be described concisely – even those that are novel and highly complex such as in cancer.

To improve human readability, reference transcripts (such as those from Ensembl, RefSeq and LRG) are referred to by their HGNC gene symbols. Although Ensembl transcript names can do this already, they are not reproducible. To overcome this issue, we made a separate stable numbering system for reference transcripts based on ascending order of ENST IDs. We call these **HGNC stable transcript IDs**.

In addition to traditional splicing events, the EDN is also directly applicable for:

- circular RNA
- splicing changes from genomic variants (when combined with HGVS[10] nomenclature), and
- protein isoforms.

A helping hand

To assist with generating, visualising and interpreting the Extended Delta Notation, we developed an R Shiny web app called the **EDN Suite**.

Currently, the EDN Suite consists of two subprograms‡:

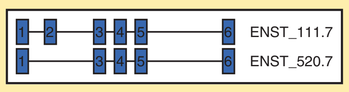
1. **EDN Automator**: generate publication-ready EDN in a single click
2. **EDN Workshop**: View transcript and protein annotation tracks with respect to the genome, including exon numbers and HGNC stable IDs. Type in a range to automatically show splicing changes in exons, protein domains and PTMs.

With easily generatable and interpretable nomenclature, the EDN paves the way for more uniform reporting standards in the future.

Tidying up messy names

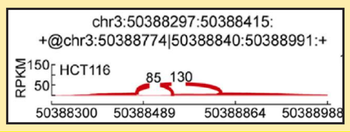
Better representation using EDN shorthand

Shah et al. (2020), *Oncogene*[6]
Format: full length isoform schematic
Type of splicing: cassette exon



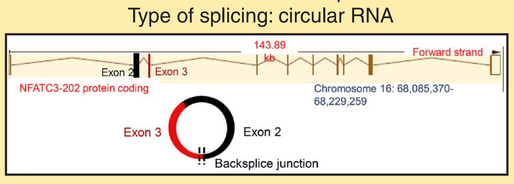
ENST_111.7 → ***PLA2G2A-enst1.7t***
ENST_520.7 → ***PLA2G2A-enst1.7t ΔE2***

Xiong et al. (2016), *Oncology Reports*[7]
Format: Sashimi plot & genomic coordinates
Type of splicing: A3SS



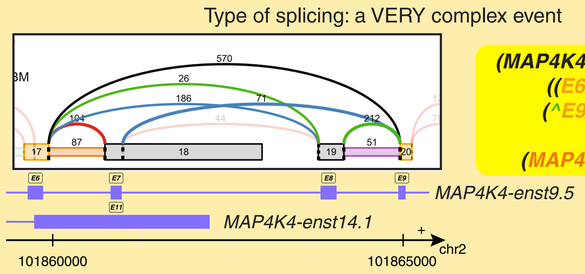
CYB561D2-enst1.5 E1^(E2/Δ66E2)

Karedath et al. (2021), *Front. Cell Dev. Biol.*[8]
Format: ensembl transcript name
Type of splicing: circular RNA



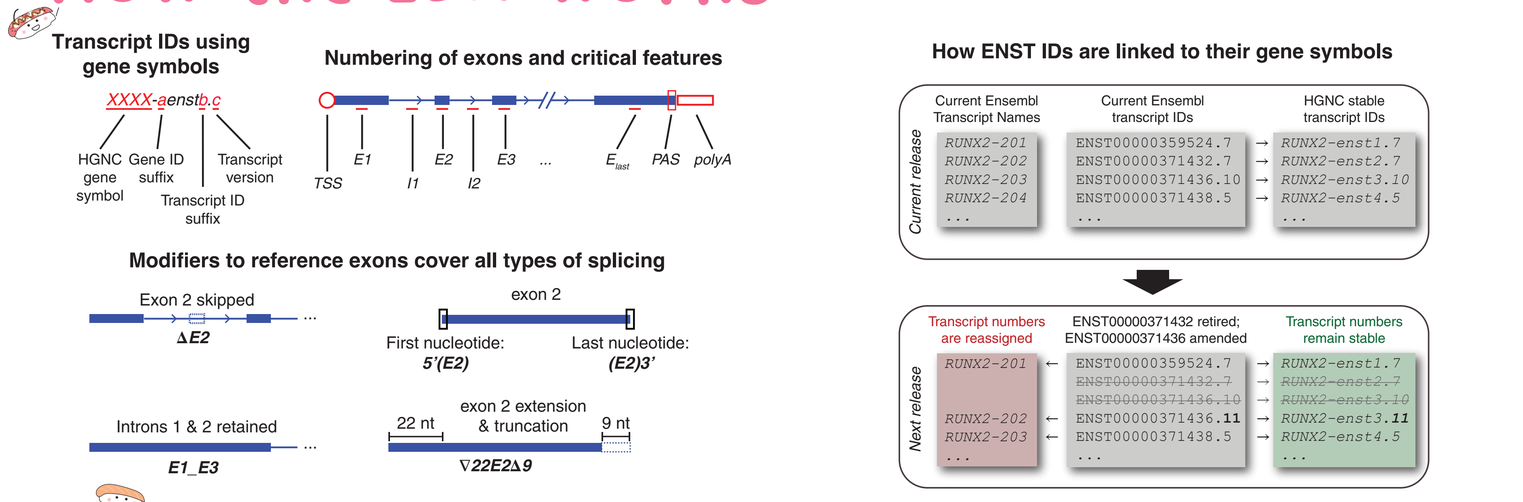
NFATC3-enst2.8 circ(E2 E3)

Wang et al. (2021), *Genome Biology*[9]
Format: LSVs from MAJIQ/VOILA
Type of splicing: a VERY complex event

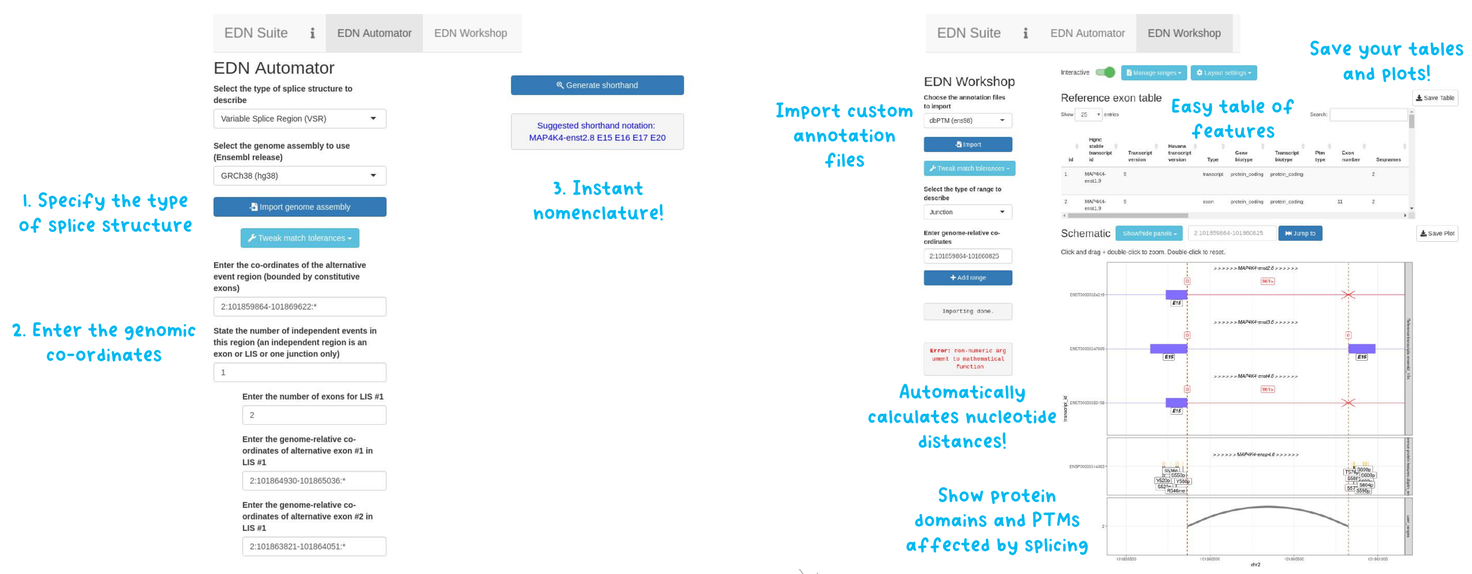


(MAP4K4-enst9.5 E6^E7^E9/((E6^E8/E6^Δ6E8)(^E9/I8 E9))/E6^E9) / (MAP4K4-enst14.1 E11)

How the EDN works



Webapp: EDN Suite



Notes

*We call the our shorthand system the "Extended Delta Notation" because we intended it to be a generalisation of the familiar Delta (Δ) convention for exon skipping.

† Other components of the code are also derived from existing naming conventions such as the AS Code (^)[11], intronic positions (+/-)[10] and the ENIGMA intronic insertion symbol nabla (∇ or ▼)[12].

‡The EDN Suite is still in development. We are adding a third subprogram called the "EDN Reverse Translate" which converts EDN to genomic co-ordinates and automatically draws a schematic. It will also allow users to check whether two different EDN representations are equivalent, which is useful for cross-referencing the literature.

References

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